

FULL GUIDE - OFFLINE DANTE XML EDITING

Dante Config Editor V3.6

Full user guide

This tool began as an attempt to provide what I personally was missing in Dante Controller: review an entire configuration quickly, correct discrepancies, and prepare a preset offline.

One overview

Latency, sample rate, network, IP, clock, and channels on one screen.

Consistent renaming

Recognized references follow renamed devices and channels.

Offline preparation

Review, merge, and edit before official validation.

Unofficial third-party tool. It does not connect to the Dante network. Always keep the original and validate the final XML in Dante Controller.

Public project: github.com/Mamat79/DanteConfigEditorV3

1. Installation and startup

Important: this is a third-party tool, not an official Audinate product. V3.6 is a development version and may still contain bugs. It edits XML files offline without connecting to a Dante network or using an Audinate API. Keep the original and validate the generated file in Dante Controller before production use.

The Windows x64 installer includes the application and the required .NET 8 runtime. A separate .NET installation is normally not required.

- The default location is Program Files, with Start menu and desktop shortcuts.
- A fresh V3.6 installation uses its own folder and shortcuts so it can coexist with stable V3.4.2.
- V3.6 upgrades the V3.5 development line when it is already installed and preserves local working data.
- The wizard offers separate active-bank and included-bank folders. DCE Generic Roles 3.6 and Yamaha QL1 + Fohhn DI4.1000 are optional, can be selected independently, and no existing folder is replaced.
- Two self-contained V3.6 DMGs are planned for Apple Silicon and Intel. They include both archives in the Machine Banks folder; the separate V3.6 bundle can coexist with V3.4.2.
- All four French and English PDFs are installed and remain available from the application.

2. Safety principles

- Work on a copy of the exported XML and use Save as.
- The guard tracks devices by stable technical identity, blocks unknown paths, and protects Dante IDs, mediaType, and instance_id.
- The destination is replaced atomically. The source and any existing destination receive a copy in DanteConfigEditor_Backups.
- A successful import into Dante Controller is the final validation before operation.

3. Open a project

Click Open XML, choose the file, then review device, TX/RX channel, and active subscription counts. XML files with a default namespace are supported. Language and theme can be changed at any time.

Screen map

The top bar opens, merges, saves, undoes, and restores the project. The Project column stays visible for counters, alerts, and search.

Screen	Main purpose
Configuration	Overview, selected device, channels, quick lists, global actions, and device table.
Patch	Tabular review and editing of Rx-to-Tx subscriptions, with filters and direct renaming.
Easy patch	Visual matrix, selection/range tools, immediate apply, and optional replacement warning.
Import / Export > Labels	JSON/CSV, DMT XLSX/ODS, A&H,, and Yamaha exchange with an import report.
Import / Export > Reports	TXT/PDF reports, TXT/CSV patchbooks, and a simple text topology.
Import / Export > Synoptic	Locations, order, visibility, zoom, reset, and SVG/PDF exports.
File health	Errors, warnings, subscription information, and review filters.
Safety and log	Validation, final report, compatibility, history, XML comparison, and user guides.
Atomic Bomb	Configurable, undoable, offline troubleshooting exercise generation.

The essentials on one screen

The Configuration page addresses the software's original need: review an entire preset quickly without opening each Dante Controller page in turn.

Spot issues Colored rows and the side banner highlight important discrepancies.	Target safely Filters, multiple selection, and locks define exactly which devices are affected.	Review Before / after lets you inspect every change before saving.
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4. Configuration page

The Configuration page combines the selected device, its channels, global actions, and the device table.

Selected device

- Apply the name, network mode, latency, and Preferred Master state together with Apply settings.
- Double-click a row or use Device details to edit IP settings, sample rate, bits per sample, TX/RX names, and subscriptions for its Rx inputs.
- A complete Device details change is grouped into one model rebuild.
- Clear actions can disconnect Rx inputs, remove subscriptions using Tx channels, or do both.
- Deleting a device also removes subscription points that reference it.

Device table

- Multiple selection defines the Selected unlocked target. The Lock column protects devices from global actions.
- Preferred Master can be toggled directly. Hide settings enlarges the table.

Search, filters, and global actions

- Search finds devices, channels, and subscription references after at least two characters.
- Quick lists filter network modes, latencies, sample rates, bits, static IPs, and Preferred Masters.
- Modified only shows changed devices; Before / after lists every difference.
- Choose all unlocked, selected unlocked, or visible unlocked devices. A preview is shown before application.

Edit a device without leaving the workflow

Device details combines the essential settings and lets you move directly to another device from the top menu.

Identity Device name, network mode, and Preferred Master.	Audio Latency, sample rate, and bits per sample.	Network Automatic or static primary IP without changing secondary interfaces.
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- The RX then TX tabs rename individual channels.
- Rx patch reviews or disconnects subscriptions received by the device.
- Apply validates the complete edit as one grouped operation; Cancel leaves the XML unchanged.

5. Navigable alerts

The Items to check banner reports mixed redundant/daisychain modes, static IPs, multiple sample rates, and multiple bit depths.

- Click Show devices, choose an alert, then review the filtered devices.
- After correcting an item, verify that the alert disappears and review File health.

6. Quick profiles

Profile	Applied settings
48 kHz / 24 bit / 1 ms	Automatic IP
48 kHz / 24 bit / 2 ms	Automatic IP
96 kHz / 24 bit / 1 ms	Automatic IP
96 kHz / 24 bit / 2 ms	Automatic IP
48 kHz / 24 bit / 1 ms / Redundant	Redundant mode and automatic IP
48 kHz / 24 bit / 1 ms / Daisychain	Daisychain mode and automatic IP

Verify that every device supports the requested sample rate, bit depth, latency, and network mode.

7. Automatic recovery

After a change, the application waits briefly and writes recovery data in the background without blocking the interface or replacing the source XML.

- When reopening the same XML, choose whether to restore or discard the previous session.
- After Save as, the new file becomes the reference for later edits and recovery data.
- The copy is deleted after saving or reverting; copies older than 30 days are cleaned automatically.

Two patching workflows

Patch remains the precise tabular editor. Easy patch adds visual selection, ranges, and immediate apply.

Patch Filter TX/RX devices, find a source, and apply or remove a subscription.	Easy patch RX on the left, TX on the right, multiple selection, ranges, and immediate apply.	Control Warning enabled by default before replacing an already-patched Rx channel.
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8. Channels and subscriptions

- TX/RX channels can be renamed individually or by range with {00}, {000}, {n}, and {device}.
- Renaming a Tx channel updates every recognized subscription alias in the project.
- In Rx/Tx lists, click the name, then press Enter to validate, Tab to validate and move forward, Shift+Tab to move backward, or Escape to cancel the edit.
- In the Easy patch matrix, click a vertical Tx label to rename it. Enter, Tab, Shift+Tab, and Escape behave exactly as they do in the lists.
- The fill handle appears only when the name ends with a number. Mic 4 and Mic 04 can be extended; Mic, Mic left, and Mic 4 main cannot.
- Fill keeps the text before the number and any leading zeroes: Mic 04 becomes Mic 05, Mic 06, and so on. Cancelling the drag changes no channel.
- Dante IDs are not renumbered. The local subscribed_device="." marker is preserved.
- The Easy patch tab shows Rx channels on the left and Tx channels on the right. Menus and arrows switch devices quickly.
- Select equal Tx and Rx counts for one-to-one mapping, or one Tx to feed several Rx channels.
- Several Tx channels to one Rx and unequal multiple selections are blocked.
- Range patching requires a first Tx, first Rx, and exact count; an incomplete range is blocked as a whole.
- A click or drag immediately applies the affected crosspoints, with no preview step.
- Clicks and drags update only the affected cells: the entire matrix is no longer rebuilt after every action.
- Selections, ranges, and PATCH 1:1 are also applied immediately.
- Warn me when the Rx channel is already patched is selected by default. Clear it only to replace a subscription without that warning.
- In the compact matrix, Rx channels are rows and Tx channels are columns. Click for one assignment, or hold and drag horizontally, vertically, or diagonally for a safe range.
- Each immediate operation remains reversible with Undo.
- In Device details, the top menu switches devices and protects unapplied changes.

Starting name	Handle	Result
Mic 4 [drag]	Visible	Mic 5, Mic 6, Mic 7...
Mic 04 [drag]	Visible	Mic 05, Mic 06, Mic 07...
Mic left	Hidden	No fill action offered

9. Add XML to project

- Devices with unique names are always imported.
- Only conflicting names are offered for automatic or manual rename.
- Imported subscriptions follow renamed imported devices.

10. IP and audio formats

- Automatic or static IP is editable per device or through a global action.
- Only the primary IPv4 interface, network=0 when available, is targeted. A secondary interface is not changed.
- DNS is not rewritten implicitly. Gateway changes only when the action provides a value.
- Sample rate and bits per sample are editable per device, globally, or through a profile.

Incorrect settings can make a device unreachable or incompatible. Verify actual hardware capabilities.

11. File health, comparison, and Import / Export

- File health combines statistics, errors, warnings, free/local subscriptions, and compatibility checks.
- XML comparison displays differences in a table.
- TXT/PDF exports include the application version and the By Mamat et ses agents signature.
- Import / Export groups Labels, Reports and patchbook, and Synoptic. The synoptic remembers locations, shows or hides devices, provides a separate preview whose zoom preserves proportions, and exports SVG or PDF; its local layout sidecar never changes Dante XML.

Exchange labels without an external template

dLive, Avantis, Yamaha CL/QL, and DMT templates are bundled with Dante Config Editor. Native export only asks for the new file name and folder.

Choose the correct format

Format	Destination	Content
Generic JSON / CSV	DCE or third-party tool	Full Unicode. Do not import into dLive Director.
DMT XLSX/ODS dLive / Avantis	dLive MIDI Tools	Direct DMT workbook; rows outside the selection are disabled.
Native A&H; CSV dLive	dLive Director	dLive [Version]/[Channels] structure and Input names.
Native A&H; CSV Avantis	Avantis Director	Avantis [Version]/[Channels] structure and Input names.
Native Yamaha ZIP CL / QL	CL/QL Editor	Complete nine-CSV package; only InName.csv receives labels.

Workflow

- Under Import / Export > Labels, choose Export labels.
- Choose TX or RX, devices, first channel, and count. A device with RX but no TX automatically switches to RX.
- Choose the native format matching the real model. Devices without channels in the selected direction cannot be checked.
- Review the preview. Enable ASCII/eight-character adaptation only when required, then choose Export.
- DCE opens Save as directly. Output is written atomically, so a failed export does not destroy an existing file.

During import, DCE reports the detected format, source version, lists, devices, channels, ignored rows, empty labels, duplicates, and warnings. Apply requires at least one error-free change. After loading the same labels again, the button intentionally remains disabled and DCE states that the labels already match.

DMT 2.14.0-RC1 JSON/CSV exports are checked with fixtures generated by the DMT exporters at commit 3c34052. XLSX/ODS support continues to target the Channels sheet from observed DMT workbooks.

Before use, always open the generated file in DMT, dLive Director, Avantis Director, or Yamaha CL/QL Editor and verify labels and the selected model.

Bundled DMT workbooks come from Tobias Grupe's MIT-licensed dLive MIDI Tools project. DMT_LICENSE.txt is included with the application.

Machine bank and generic roles

These functions edit offline preset roles, not real devices. DCE removes hardware identifiers from the source instance and does not claim to create a Dante device identity.

Duplicate a device

- Select a device, choose Duplicate, and provide a unique role name.
- TX/RX labels may be retained. Network data, settings, flows, Preferred Master, and subscriptions are disabled by default.
- The source is unchanged. The copy is added as one undoable operation and receives a DCE session identity that is never serialized.

Save and share a template

- Choose Save to machine bank and enter manufacturer, model, category, description, tags, and generic labels.
- An optional PNG, JPEG, or WebP image is copied into the model folder; no fragile external path is kept.
- The default bank is Documents/Dante Config Editor/Machine Bank. You may choose, open, copy, or place it in a synchronized folder.
- Export bank creates a verified *.dce-bank.zip archive. Import bank requires a new or empty folder and never replaces existing data.
- GitHub banks opens the public V3.6 catalog. DCE Generic Roles 3.6 provides generic 8x8 and 32x32 test roles. The Yamaha QL1 + Fohhn DI4.1000 community bank provides two sanitized illustrated templates. These banks contain no hardware identity, network data, or subscription.
- Administration supports search, filters, edit, duplicate, confirmed delete, and model ZIP import/export.

Add a template to the project

- Choose Add from bank, then configure the new name, labels, and only the options you explicitly need.
- The inserted instance is independent from the template. Editing either one never changes the other.
- DCE checks model version, checksum, channel counts, namespace, and preset version before a transactional insertion.

Experimental new project

- New project writes a minimal 3.0.0 structure, either empty or with one initial role from the bank.
- An existing file is never overwritten silently. Writing uses a temporary file followed by validation and atomic replacement.
- Reopen the XML in DCE, review File health, then perform a mandatory manual import into Dante Controller.

Technical logs are available from Safety and log. They explain import, validation, bank, and export failures without changing the project.

Review before saving

The final two pages explain anomalies, produce reports, and help verify that only the intended changes will be kept.

File health Errors, warnings, mixed audio formats, static IPs, and local subscriptions.	Safety and log Pre-save summary, XML compatibility, reports, history, and user guides.	Atomic Bomb Separate tab, configurable categories, and three required confirmations.
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12. Atomic Bomb: create an exercise

- Open the Atomic Bomb tab after Safety and log. Clear the categories you want to spare; all are selected by default. Three confirmations then describe the consequences before any change.
- The in-memory copy receives unique mythological, audio-themed, or playful names plus a mixture of subscriptions, network modes, Preferred Master states, latencies, sample rates, encodings, and primary IP settings.
- Technical identifiers, namespaces, DNS, gateways, and secondary interfaces remain protected.
- The summary displays the scenario seed. The entire operation is one undo step and the source file is never overwritten.
- Use Save as to provide the trainee preset, then verify its import in the appropriate official Dante tool.

This mode is only for offline training. It does not alter any device or communicate with the Dante network.

13. Save and final validation

Use Save as. The temporary XML is reloaded, protected changes are checked, and the destination is replaced atomically. A failure before replacement leaves the previous destination intact.

Check	Recommended action
Items to check	Open affected devices and explain or correct every unexpected difference.
Modified only	Verify that only the intended devices appear.
Before / after	Review every changed setting, channel, and subscription.
Dante Controller	Import the file into a working copy before any field operation.

14. Regression tests

The V3.6 suite runs 272 Core/Windows tests and 20 headless Mac tests. Coverage includes XML guards, save and recovery, IPv4 interfaces, subscriptions, large presets, duplication, the machine bank, experimental project creation, DMT formats, import reports, synoptic export, Atomic Bomb, Easy patch, and translation consistency.

15. Known limitations

- No real-time Dante control and no communication with devices.
- No Audinate SDK/API and no proprietary protocol bypass.
- Compatibility depends on the XML structure; only an official import confirms the final file.
- Undo keeps at most 10 states to limit memory use.
- The matrix displays only the two selected devices to preserve performance on large presets.
- Mac DMGs are ad hoc signed but not notarized; first launch may require right-clicking the application and choosing Open.
- The Windows Easy patch tab is not reproduced identically on Mac, which keeps the Avalonia visual patch workshop.
- Duplicate Tx names are ambiguous in Dante subscriptions and must be renamed before using Easy patch.
- Native workbooks match observed DMT 2.13.0 templates and the supplied dLive, Avantis, CL5, and QL5 examples; DMT 2.14.0-RC1 JSON/CSV is tested separately.
- Generic roles duplicated or added from the bank carry no hardware instance_id/device_id. Only an actual Dante Controller import can confirm their use with a given version.
- New project writes an experimental minimal 3.0.0 structure. It must be imported and reviewed in Dante Controller before use.

16. Help and information

Quick start and Full guide automatically open the French or English PDF for the active language. Public project:
github.com/Mamat79/DanteConfigEditorV3 - Credit: By Mamat et ses agents.